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Full Length Article

AI-induced job impact: Complementary or substitution? Empirical insights and sustainable technology considerations

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ABSTRACT

This study utilizes 3,682 full-time workers to examine perceptions of AI-induced job displacement risk and evaluate AI's potential complementary effects on labor. By distinguishing between these variables, the research identifies which worker characteristics and industries are most affected by AI. Employing an extended ordered probit model, the results show that those with higher perceived risks are mainly female, older, and more educated. Workers with frequent internet usage and remote work are more concerned about AI-induced job risks. Sector-wise, entry-level employees perceive the least risk, whereas manufacturing professionals and senior service sector employees are more apprehensive. Contrary to the mainstream assumption that AI causes unemployment, this study identifies a potential complementary effect on job tasks, suggesting that AI can enhance and support human labor rather than replace it. This inference is based on observed worker perceptions and the increased use of AI tools in tasks that complement human skills. Moreover, AI's integration contributes to sustainable practices and entrepreneurial opportunities, enhancing business innovation and environmental efficiency. Workers increasingly view AI as a value-added element in their jobs rather than a threat, particularly in sectors focusing on sustainable development and green technologies.

Introduction

Launched in 2022, ChatGPT, a GPT-4-based large language model, rapidly gained prominence. Utilizing supervised and reinforcement learning, it offers near-human conversation and complex task-solving abilities. While well-received in economic and social domains, Artificial Intelligence (AI) also rekindles concerns about labor displacement on a larger scale and across a broader scope similar to earlier automation technologies. Additionally, AI's role in promoting sustainable technologies and entrepreneurial ventures has become increasingly significant, driving innovation and efficiency in various sectors.

Before the AI surge, automation had been a critical focus in labor market research. Frey and Osborne (2017) conducted a seminal study revealing that 47 % of 702 U.S. occupations were at high risk of automation, particularly in transportation and unskilled production sectors. This study highlighted the potential for job displacement due to automation, setting a benchmark for further research on technology's impact on labor markets. In contrast, Badiuzzaman and Rafiquzzaman (2020) examined the threat posed by automation and robotics, specifically on unskilled and lower-skilled labor in highly populated countries, emphasizing the vulnerability of these worker groups. Their findings underscored the socio-economic risks associated with rapid technological advancements. However, AI's ability to foster sustainable practices and create new entrepreneurial opportunities presents a nuanced impact on the labor market, contributing to economic resilience and environmental sustainability.

Dwivedi et al. (2021) reviewed the emerging challenges and opportunities AI brings, emphasizing the multidisciplinary perspectives required to understand AI's impact on employment. This study contributed to a broader understanding of how AI poses risks and offers avenues for new job creation and enhanced productivity. Schmidpeter and Winter-Ebmer (2021) explored the role of labor market training in mitigating the adverse effects of automation on employment, highlighting the importance of continuous skill development in maintaining workforce relevance.

Nilsson (1984), DeCanio (2016), and Sorells (2018) delved into the broader implications of technology on labor. Nilsson (1984) argued against the need to curb intelligent production due to unemployment fears, suggesting that innovative technologies could enhance productivity and living standards. DeCanio (2016) examined whether robots

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and humans are complements or substitutes, concluding that the relationship depends significantly on the implementation context. Sorells (2018) focused on the potential rate and extent of job displacement caused by workplace automation, providing a nuanced perspective on the immediate versus long-term impacts.

Autor (2015), however, presented a historical analysis, contending that past waves of automation have not resulted in mass unemployment. Autor acknowledged the disruptive potential of AI but argued that automation historically leads to job transformation rather than elimination. This perspective is crucial in understanding that the effects of AI and automation on labor markets are different. While automation typically replaces routine tasks, AI can complement human labor, enhancing job roles and creating new opportunities for skilled workers. This study builds on these diverse insights to examine AI's nuanced impact on job displacement in Taiwan. By evaluating workers' perceptions and their engagement with AI tools, the study highlights the potential complementary nature of AI in modern labor markets, suggesting that AI can work alongside human labor to enhance productivity and innovation.

Technically, AI and automation differ fundamentally. AI mimics human decision-making and learns through machine and deep learning, enabling it to adapt and tackle complex issues like speech and image recognition. Automation, in contrast, operates on pre-set rules and is generally non-adaptive, best suited for more straightforward, repetitive tasks. While AI prioritizes adaptability and intelligence, automation centers on executing fixed, predetermined tasks.

Extending the perspective to the impact on the labor market, AI demands highly skilled roles like data scientists, transforms industries such as healthcare and finance, and potentially creates new job types. However, it also sets a high entry barrier. These factors prevent workers from entering the market, especially those needing more technical expertise in the new job types that have been opened due to AI technology. Conversely, automation mainly impacts low-skilled, repetitive jobs, leading to job declines in areas like manufacturing (Frey & Osborne, 2017). It may also generate jobs in system operations and maintenance, allowing for a geographically diverse job distribution (Bessen, 2018). AI and automation require ongoing skill development from workers, resulting in structural labor market shifts (Acemoglu & Restrepo, 2018).

AI's impact on the labor market is multifaceted. Agrawal et al. (2019) emphasize that AI's effects cannot be determined solely from factory automation literature because AI has broader implications beyond factory settings. It impacts various sectors differently, presenting both positive and negative implications. For instance, AI can enhance productivity and create new job opportunities in some industries while displacing others. It highlights the necessity of examining AI's impact across diverse contexts to understand its full implications. Given the heightened global AI discourse in 2022, this study uses Taiwan's 2022 Digital Development Survey data to examine full-time workers' views on AI-induced job displacement. It will also investigate AI's potentially complementary or substitutive relationship with labor, i.e., AI may enhance and support human labor rather than replace it.

This study diverges from the existing literature in three key aspects. First, it uses national-level data collected at the height of AI discourse to evaluate workers' perceptions of AI-induced job displacement, identifying which worker characteristics and industries are most affected. Second, our empirical model simultaneously explores whether AI acts as a labor substitute or a labor complement at varying degrees of perceived job displacement. Finally, given that AI's influence on labor is still in its infant stage, this study offers timely insights into worker attitudes, informing future business strategies for navigating AI integration.

Analyzing 3682 Taiwanese worker samples via an extended ordered probit model, this study finds that those perceiving higher AI displacement risks are mainly female, older, highly educated, frequent internet users, and remote workers. Professionals in manufacturing, senior executives, and service industry managers are particularly concerned about AI's impact. Contrary to what has happened in the past concerning automation, those workers just starting in their careers perceive the lowest risk of losing their jobs to AI. Importantly, AI currently complements rather than substitutes for labor tasks, challenging mainstream claims of AI-induced mass unemployment. Utilizing 2022 Taiwanese official data, this study provides a nuanced, empirical view of the global AI landscape, offering valuable insights for academia, public policy, and general perceptions of AI's labor market impact.

Beyond this introduction, the paper is structured as follows: Section 2 reviews the literature on AI and labor; Section 3 details data sources; Section 4 explains the empirical methodology; Section 5 discusses analytical results; and the final section concludes, suggesting avenues for future research.

Literature review

AI has advanced rapidly, eliciting wide-ranging socio-economic debates. Nilsson (1984) argues against the need to curb intelligent production due to fears of unemployment, suggesting that it can enhance global productivity and living standards by automating mundane tasks, thereby allowing human workers to focus on more complex and rewarding activities., emphasizing that it can boost worldwide productivity and improve living standards. Verma and Singh (2022) emphasize AI's potential for organizational competitiveness and employee development. While both agree on AI's productivity, they differ in terms of its social impact. Dwivedi et al. (2021) characterize AI as mimicking human cognition, underscoring its capacity for autonomous learning and multi-domain applications.

The relationship between workplace automation and labor productivity growth in the context of AI-induced job displacement risks is critical to understanding the broader impact of AI on labor markets. Rivera-Hernaez et al. (2019) highlight that AI and robotics solutions can augment professional capabilities by automating routine tasks and enabling workers to focus on higher-order functions, increasing overall productivity. Similarly, Tomé et al. (2020) reveal that consumers and producers in Portugal and Spain perceive AI as a tool that can enhance job efficiency and create new opportunities rather than merely displacing jobs. Zaldo-Santamaria et al. (2020) provide a systemic view through system dynamics, demonstrating how digitalization and AI contribute to competitiveness and employment by fostering new roles and improving job quality. These studies collectively underscore the complementary nature of AI, suggesting that while AI can automate specific tasks, it simultaneously creates new opportunities for human labor by enhancing professional capabilities and fostering innovation. This dual nature of AI's impact is further supported by Nica et al. (2023a, 2023b) and Lăzăroiu and Rogalska (2023), who analyze the integration of AI in workplace tracking systems and the resulting labor productivity growth. The complementary relationship between AI and labor highlights the importance of adaptive strategies and continuous skill development to manage AI-induced changes effectively.

The academic discourse also offers a range of complementary perspectives. DeCanio (2016) and Bruun and Duka (2018) highlight AI's role in mimicking or exceeding human cognition, often referring to these systems as robots. Kaplan and Haenlein (2020) extend the discussion, stressing AI's versatility in data interpretation and learning. Guliyev (2023) notes AI's significant impact on human life and work, while Bordot (2022) further explores AI's adaptability, particularly demonstrating exceptional problem-solving capabilities in contexts requiring the encoding of tacit knowledge.

Research findings on the interplay between AI and the labor market have been varied and inconclusive. Nilsson (1984) and Mutascu (2021) highlight AI's dual effects, pointing to economic growth and job transformation. Bruun and Duka (2018) suggest that while increasing productivity, AI has significantly contributed to job displacement, advocating for policy measures like universal basic income to mitigate social instability. Bordot (2022), through a comprehensive analysis of

OECD countries, finds a positive correlation between the rise in industrial robots and AI-related patents and increased unemployment rates, especially among medium-skilled workers, indicating that technological advancements might exacerbate labor market polarization. DeCanio (2016) suggests that AI might have a depressive effect on worker wages, which is echoed by Mutascu and Hegerty (2023), who associate AI with rising unemployment.

Jarrahi (2018) and Nguyen and Vo (2022) offer a more optimistic view, emphasizing AI's role in enhancing decision-making and job value. Song et al. (2022) believes that AI and human labor can collaboratively deliver superior services. Kaplan and Haenlein (2020) and Mutascu (2021) agree that AI mainly displaces low-skilled jobs. Notably, AI's impact extends beyond labor structure to psychological pressures. Verma and Singh (2022) and Li et al. (2023) stress that AI incites fears of worker obsolescence and reduced learning motivation, particularly severely impacting specific groups.

In distinguishing AI from traditional automation, Nilsson (1984) argues that AI performs tasks initially done by humans more cost-effectively, potentially raising unemployment. Zhou et al. (2019) observe that machines are now incorporating 'intelligence,' resulting in the transformation of production processes. DeCanio (2016) suggests that the impact of technology on employment or wages depends on the extent to which it can be substituted for labor. Agrawal et al. (2019) warn against extrapolating AI's labor impact solely from factory automation studies, as AI's effects can be dual. Frey and Osborne (2017) and Dwivedi et al. (2021) agree that automation mainly impacts low-wage and low-education workers, potentially reducing labor hours and increasing unemployment. Kaplan and Haenlein (2020) suggest AI could positively impact labor through real-time monitoring to improve efficiency. Overall, the literature underscores technology's nuanced and multi-dimensional effects on the labor market.

The scholarly consensus suggests that AI poses substitution risks primarily for low-skilled and repetitive jobs (DeCanio, 2016; Frey & Osborne, 2017; Verma & Singh, 2022; Dwivedi et al., 2021). In the manufacturing and transportation industries, white-collar roles are increasingly facing AI-driven displacement (Jarrahi, 2018). This effect varies by region and demographic, with pronounced impacts in rural and urban China (Zhou et al., 2019) and among younger, less-educated workers (Bordot, 2022).

However, the impact of AI on the labor market is not exclusively negative. Agrawal et al. (2019) and Guliyev (2023) suggest that AI could positively impact certain professions, varying by industry and application. Inflation rates might also influence AI's employment effects (Mutascu, 2021; Nguyen & Vo, 2022). While AI poses risks to some jobs, it also offers the potential for new employment and increased productivity. Future studies should assess these dual impacts across socio-economic groups. Nguyen and Vo (2022) propose a holistic strategy that balances economic growth and social innovation. Their approach includes developing new policies and initiatives that support workforce transition and financial resilience.

Drawing on existing literature, AI has rapidly evolved over recent decades, impacting not just economic growth but also labor markets and social structures. Academic views on AI are diverse, covering its definition and applications from human-like cognition to various practical implementations. While AI promises to boost global productivity and job creation, it also aggravates social and economic disparities, particularly in employment and wage conditions. Consequently, a more comprehensive evaluation of the various impacts of AI is essential. Based on this foundational understanding, this study aims to analyze AI's perceived job displacement risks for human labor.

Data

The empirical data for this study is from the 2022 Digital Development Survey (DDS) released by Taiwan's Ministry of Digital Affairs (MODA) in May 2023. Conducted annually since 2001, the DDS includes up-to-date questions related to information and communication technology and internet development for each year, aimed at capturing current societal concerns regarding digital life issues.

In the 2022 survey, participants assessed the likelihood of AI replacing their current jobs, with options including "highly likely," "somewhat likely," "not very likely," and "not at all likely." To keep the focus on AI's impact on the full-time labor force, only respondents who conformed to Taiwan's labor force definition, i.e., those aged 16–65, were considered. Subsequently, data from 3682 workers was analyzed.

Table 1 indicates varying perceptions among Taiwanese workers on the likelihood of AI replacing their jobs, stratified by industry. Manufacturing workers report higher rates of "highly likely" and "somewhat likely" responses, while those in the service sector lean towards "not very likely" and "not at all likely." These patterns suggest industry-specific trends in AI labor displacement. The manufacturing sector's routine, repetitive tasks, and capital-intensive nature make it more prone to AI adoption. Conversely, the service industry's job complexity and emphasis on human interaction make it less susceptible to AI-driven displacement, particularly when consumers favor human services. Overall, manufacturing jobs appear more vulnerable to AI and automation.

Regarding occupational levels, technical personnel most frequently indicated a high likelihood of AI displacement, dominating the "highly likely" and "somewhat likely" categories. In contrast, professionals often chose "not very likely," and senior executives and managers primarily opted for "not at all likely." Entry-level employees had higher rates of "not very likely" and "not at all likely" than technical and professional staff. These findings suggest varying AI displacement risk perceptions across occupations. Those in technical roles, often routine, feel their jobs are most replaceable, while professionals endowed with unique skills feel secure. Those occupying Senior roles, who tend to focus on less automatable tasks like strategy and human resources, express less concern. Despite their often-repetitive nature, entry-level jobs are less targeted for AI displacement due to the essential human interaction and production involved.

Based on the analysis in Table 1, the manufacturing sector is more susceptible to AI disruption than the service sector due to its work's repetitive and predictable nature. On an occupational level, technical personnel are the most concerned about being replaced, primarily because their tasks are easily automatable. Conversely, professionals, senior executives, and managers feel relatively secure at this stage due to the complexity of their roles. These findings reveal significant variations across different industries and occupational levels in facing the challenges posed by AI.

Empirical method

Given that in the DDS survey, workers' perceptions of AI-induced job displacement risk and evaluations of AI's complementary effects on labor, this study designates these perceptions as ordered variables ($AIdisplace_i$).

$$AIdisplace_i = \begin{cases} 1 & \\ 2 & \\ 3 & \\ 4 & \end{cases} \quad if \quad \begin{matrix} Not\ at\ all\ likely \\ Not\ very\ likely \\ Somewhat\ likely \\ Highly\ likely \end{matrix} \tag{1}$$

Since employees' views are shaped by personal traits, job factors, and sector connections, AI displacement perception and complementary effects are treated as endogenous variables. Drawing from the approach used by Rivera-Hernaez et al. (2019), which highlighted the importance of differentiating between tasks that can be automated and those that require human intervention, our model also categorizes job tasks based on their susceptibility to AI automation. This differentiation allows for a more nuanced analysis of AI's impact on labor. Similarly, Tomé et al.

Table 1
The proportion of Taiwanese workers perceiving job displacement by AI.

	Industry		Occupation			
	Manufacturing	Service	Senior executives and Managers	Professionals	Technicians	Grassroot staff
Highly likely	12.1212 %	8.8196 %	8.6680 %	8.6419 %	10.4166 %	10.2921 %
Somewhat likely	23.1060 %	17.8382 %	16.2790 %	18.1657 %	22.6190 %	18.2324 %
Not very likely	41.9191 %	47.6790 %	44.6088 %	49.3827 %	41.3690 %	43.6546 %
Not at all likely	22.8535 %	25.6631 %	30.4439 %	23.8095 %	25.5952 %	26.8207 %

(2020) emphasized the importance of understanding consumer and producer perceptions in AI integration. This informs our decision to include worker perceptions as a critical variable in our model. Zaldo-Santamaria et al. (2020) provided insights into the systemic impacts of digitalization, suggesting that AI's effects on competitiveness and employment should be viewed through a holistic, dynamic lens. These perspectives have been incorporated into our extended ordered probit model to capture the multifaceted nature of AI's impact on labor markets. The study's regression model is structured accordingly, as follows:

$$\begin{aligned} AIdisplace_i = {} & \alpha_0 + \alpha_1 Sex_i + \alpha_2 Age_i + \alpha_3 Education_i + \alpha_4 iUsage_i + \alpha_5 Remote_i \\ & + \alpha_6 ManufacturingSenior_i + \alpha_7 ManufacturingProfessionals_i \\ & + \alpha_8 ManufacturingTechnicians_i + \alpha_9 ManufacturingWorker_i \\ & + \alpha_{10} ServiceSenior_i + \alpha_{11} ServiceProfessionals_i \\ & + \alpha_{12} ServiceTechnicians_i + \alpha_{13} ServiceWorker_i + e_i \end{aligned} \tag{2}$$

The risk of AI-induced job displacement is shaped by variables like gender (Sex_i), age (Age_i), and education level ($Education_i$). For example, for gender, traditional gender roles often steer women into jobs that are more easily automatable, while men gravitate towards technical STEM fields. In particular, women's soft skills, like emotional intelligence and people skills, are a type of warranty against AI displacement. For age, younger workers, despite their adaptability and modern skills, such as digital literacy and critical thinking, may be vulnerable due to their entry-level positions. Older workers, unfamiliar with new technologies such as AI, may need to work on a match between skills and technology if they continuously learn. Finally, higher education levels are generally associated with specialized skills and job complexity, reducing susceptibility to AI displacement.

The influence of internet usage frequency ($iUsage_i$) and remote work ($Remote_i$) on susceptibility to AI automation is multifaceted. High-frequency internet users typically adapt better to tech-driven work environments, excelling in information processing and self-guided tech learning, thereby decreasing their risk of AI displacement. Remote jobs, often more knowledge- or skill-intensive, are generally less automatable due to their specialized requirements. Nonetheless, even in remote settings, more straightforward tasks remain at risk. Across both contexts, individual susceptibility to AI largely hinges on performance in specific roles and functions.

Across both contexts, individual susceptibility to AI largely hinges on performance in specific roles and tasks. Due to AI's rise, different industries and job levels exhibit varied risks and opportunities. AI chiefly targets repetitive tasks in manufacturing but generates new roles, boosting efficiency and demand for specialized skills (Frey & Osborne, 2017). However, the service sector exhibits more diversity; in high-end services requiring deep human interaction, AI is a tool to enhance efficiency rather than entirely replace human workers (Kaplan & Haenlein, 2020). Senior executives and managers remain pivotal in strategy and networking, with AI aiding in analytics and forecasting. Due to their specialized expertise, professionals face a lower risk of AI displacement, although they are increasingly integrating AI to improve efficiency. Lower-level and technical staff must continually upskill, especially in automatable areas (Bessen, 2018).

Besides examining workers' perceptions of AI-induced job displacement, this study also aims to gauge the impact on work completed via computers or the Internet. As AI advances, workers increasingly depend on digital technologies for tasks and interaction, elevating their computer and Internet usage time. AI further amplifies this trend by enhancing digital collaboration, thus heightening dependency on these platforms.

From the literature review, prior studies have deeply analyzed AI's impact on employment. So, subsequent empirical results indicate an 'increase' in the proportion of work conducted using computers or the internet due to AI; this can be interpreted as a 'complementary effect' between AI and human labor. In this scenario, AI not only refrains from directly replacing human resources but also serves as a tool or auxiliary system that assists workers in completing tasks more efficiently. This complementary relationship often leads to an increase in productivity. It may even give rise to new occupational roles or specialized fields, requiring workers to acquire new or more advanced skills.

Alternatively, if empirical results indicate a 'decrease' in computer or internet-based work due to AI, this usually suggests a 'substitution effect.' AI may have supplanted specific tasks in this context, lessening worker reliance on these platforms. Such substitution often targets repetitive or low-skilled roles, partially replacing human labor. This study aims to elaborate on AI's impact on labor market structure and worker skill demands through these effects, providing a more nuanced understanding.

For analytical methods, this study uses Stata's 'eoprobit' package. An ordered probit model serves for empirical validation, estimating Eq. (2). An extended ordered probit model is then applied to assess AI's complementary or substitutionary effects on computer- or internet-based work. This model's strengths include its suitability for ordered data variables and effective heterogeneity control, resulting in more accurate estimates.

Empirical results

Using an ordered probit model, Table 2 presents empirical results on workers' perceptions of AI-induced job displacement. Females, older workers, and individuals with higher levels of education are more likely to perceive a higher level of risk. Traditional gender roles lead females to careers like interpersonal interaction and administration, which, despite possessing unique human elements, include tasks like data entry and basic customer service that are susceptible to AI displacement. The media portrayal of AI as male-dominated and with limited opportunities for women in STEM also heightens their concerns about AI-related job displacement.

Older workers exhibit distinct concerns about AI-related job displacement. Often needing more exposure to modern AI content like systems and coding, they face a skills gap with the demands of newer jobs. Career changes or skill acquisition are more costly than younger workers. In a media landscape often highlighting potential mass unemployment due to AI, these older workers, generally bearing greater economic responsibilities, feel heightened concern and stress about possible job loss.

Contrary to general expectations, highly educated workers are more concerned about AI replacing their jobs. While traditional views suggest that specialized skills and complex roles make them less vulnerable to AI, the study finds the opposite. These workers often work in specialized, technical fields that overlap significantly with current AI development trajectories. In cases where AI can perform efficiently and specialize,

Table 2
Determinants of Taiwanese workers' perceptions of job displacement by AI.

	Perceptions of job displacement by AI
Sex	−0.1062***
	(−6.45)[a]
Age	0.0024***
	(2.60)
Years of education	0.0115***
	(8.17)
High Internet usage	0.2280***
	(13.17)
Remote work	0.2429***
	(11.16)
Senior executives and Managers in the manufacturing industry	0.2322***
	(4.94)
Professionals in the manufacturing industry	0.3366***
	(6.62)
Technicians in the manufacturing industry	0.3297***
	(7.16)
Entry-level worker in the manufacturing industry	0.1287***
	(4.97)
Senior executives and Managers in the service industry	0.3333***
	(8.64)
Professionals in the service industry	0.2536***
	(8.75)
Technicians in the service industry	0.2688***
	(6.63)
Entry-level worker in the service industry	0.1951***
	(8.79)
Wald χ^2 (p-value)	7903.43 (0.00)
Number of obs.	3682

a. t value: b. ***represent statistical significance at the 1 %.

these educated workers exhibit increased concerns about job security.

The study finds positive correlations between high internet usage and remote work and perceived job vulnerability to AI. A plausible explanation is that frequent internet users are likely more aware of AI's labor market impact, heightening their concerns. Moreover, since these workers engage in digitized tasks, they feel their roles are more susceptible to automation. This fluid job nature makes them sensitive to job stability, while their dependence on technology keeps them alert to emerging technologies, further fueling their concerns.

This study highlights the differing concerns regarding job displacement caused by AI across different industries and job levels. Manufacturing professionals and technology know more about AI displacement and entry-level workers. In contrast, the service sector sees heightened concern among senior executives and managers, outpacing entry-level employees. These results underscore divergent attitudes toward AI-related labor shifts in the manufacturing and service sectors.

In manufacturing, professionals and technicians are more susceptible to AI threats due to their specialized, standardized workflows, which often incorporate advanced robotics and AI (DeCanio, 2016). Conversely, entry-level workers in this sector are less replaceable by AI for three reasons. First, their tasks demand flexibility and improvisational skills, areas where current AI needs to be improved (Acemoglu & Restrepo, 2018). Second, even in automated environments, human oversight is necessary for monitoring, maintenance, and troubleshooting, requiring situational awareness—an attribute AI currently lacks (Brynjolfsson & McAfee, 2014). Finally, entry-level roles involve complex interpersonal interactions, encompassing variables that AI cannot fully replicate (Autor, 2015). Senior executives and managers responsible for strategic decision-making and human resource management in the service sector express elevated concern about AI replacing their complex judgment capabilities. Professionals and technicians also perceive threats from advancing AI and big data analytics. However, entry-level employees who primarily engage in client interactions face a lower risk of AI-induced displacement. These findings emphasize that concerns about AI-related job losses vary by industry and role, requiring nuanced analysis (Frey & Osborne, 2017).

Senior executives and managers responsible for strategic decision-making and human resource management in the service sector express elevated concern about AI replacing their complex judgment capabilities. Professionals and technicians also perceive threats from advancing AI and big data analytics. However, entry-level employees who primarily engage in client interactions face a lower risk of AI-induced displacement. These findings emphasize that concerns about AI-related job losses vary by industry and role, requiring nuanced analysis.

In conclusion, attitudes towards the risk of job displacement caused by AI vary due to various factors. Women, the elderly, and individuals with higher levels of education are mainly apprehensive due to social roles, digital divides, and job vulnerabilities. Regular remote internet users are similarly wary. Professionals, technicians, and senior service roles in manufacturing perceive more significant AI threats, whereas entry-level workers feel secure. Such concerns differ noticeably across industries and job levels.

In our investigation, Table 3 evaluates AI's potential effects on labor. The numbers in the table depict the odds ratio compared to those believing AI replacement is "not at all likely." The odds ratios for those considering it "highly likely, " "somewhat likely, " and "not very likely" are 2.8908, 2.0394, and 1.1219, respectively. This result shows that these workers use computers or the Internet for their jobs 2.8908, 2.0394, and 1.1219 times more often than the control group. Data from the 2022 Taiwan sample, within the backdrop of global AI advancements, indicates AI complements rather than replaces workers' tasks.

The data from Table 3 contradicts popular social and media narratives about AI-induced job losses. One possible explanation for this discrepancy is that workers have developed a certain level of technical adaptability, enabling them to adjust to and integrate into highly technical work environments and collaborate more effectively with AI-related tools. This finding aligns with Rivera-Hernaez et al. (2019), who found that AI and robotics solutions augment professional capabilities rather than simply displacing them. Furthermore, Tomé et al. (2020) observed that both consumers and producers perceive AI as enhancing job efficiency and creating new opportunities, which supports our findings that AI can complement human labor. Additionally, Zaldo-Santamaria et al. (2020) highlighted the role of digitalization in enhancing competitiveness and employment, suggesting that AI's integration into the workplace can lead to job transformation and new role creation rather than outright displacement. Workers may also overestimate their risk of AI replacement, spurring them to adopt new technologies more proactively.

From a corporate standpoint, encouraging employee use of advanced technology aligns with industry trends and can be facilitated by increased training resources. This strategy significantly enhances the volume of tasks carried out by employees using computers or the Internet, easing their concerns about being replaced by AI. These factors collectively indicate a complementary, rather than a substitutive, relationship between AI and labor, offering valuable insights into AI's role in the job market.

Table 3
Impact of AI job displacement perception on computer or internet work.

	Proportion of work requiring computers or the internet to be completed
Perceptions of job displacement by AI	
Not very likely	1.1219***
	(56.92)[a]
Somewhat likely	2.0394***
	(78.04)
Highly likely	2.8908***
	(77.51)

a. t value: b. ***represent statistical significance at the 1 %; c. "Not at all likely" is the control group.

Discussions and conclusions

Discussions

When AI and human labor synergize in the workplace, businesses and employees should strategize to optimize this trend's benefits. Companies should offer gender- and age-sensitive training to better integrate AI with human tasks, addressing the specific concerns of female and older workers. In predominantly automated sectors like manufacturing, fostering human-machine collaboration can improve efficiency and mitigate fears of total human job replacement. Furthermore, AI-driven innovations in sustainable technologies can significantly increase environmental efficiency and green entrepreneurship, positioning businesses to thrive in a sustainability-focused market.

Firms can ease job security fears for highly educated employees through specialized AI training. Open communication during AI's development and deployment phases can also reduce anxieties and uncertainties about change. Considering the varying impact and acceptance of AI across different industries and differing job levels, companies should offer tailored solutions to meet the specific needs of each sector, with a focus on sustainable practices and entrepreneurial growth.

For workers, continual education is crucial, especially for older or highly educated individuals. Given that the changes brought about by AI may induce psychological stress, the provision of mental health support and resources would be highly beneficial. As digitalization and cybersecurity grow in importance, becoming more aware of safety concerns is vital. Developing multi-skilled and cross-disciplinary competencies and considering diverse career paths will help maintain competitiveness in a fast-changing work landscape while contributing to sustainability and innovation.

The wave of AI will continue to grow and permeate all industries and occupations. Both firms and laborers need to adopt proactive, transparent, and diversified strategies to address the challenges and opportunities brought about by AI. This issue goes beyond technical and economic considerations, encompassing social, cultural, and psychological factors. It demands comprehensive and holistic solutions.

Conclusions

Using the extended ordered probit model, this study delves into workers' perceptions of AI-induced job displacement, yielding several notable findings. Contrary to prevailing beliefs, highly educated workers are more concerned about AI replacement, likely due to the alignment between AI advancements and their professional fields. Similarly, females and older workers perceive higher risks due to social roles, digital divides, and job transition costs. Regarding work nature, remote and frequent internet users feel more threatened, likely due to their higher exposure to AI's workplace impact and an increased reliance on new technologies. Industry-level findings also indicate significant variations in how AI is perceived, particularly as a threat across occupational levels within the manufacturing and service sectors.

AI is inferred to have a complementary relationship with labor based on workers' perceptions and increased digital task engagement, enhancing job roles and fostering sustainable entrepreneurial growth. This supports Rivera-Hernaez et al. (2019), who demonstrated that AI and robotics can enhance professional capabilities, and Tomé et al. (2020), who found that AI integration is perceived positively by both consumers and producers. Additionally, Zaldo-Santamaria et al. (2020) underscored the systemic benefits of digitalization on competitiveness and employment, suggesting that AI's integration can lead to new job creation and enhanced job quality. These findings collectively highlight the potential for AI to support and augment human labor, leading to more efficient and innovative work environments. Policymakers should, therefore, consider strategies that leverage AI's complementary capabilities to promote economic growth and job sustainability. This perspective aligns with findings that suggest AI can support and augment human labor, leading to more efficient and innovative work environments.

Comparing these results with previous studies, it becomes evident that AI's impact on labor markets is multifaceted. For instance, Frey and Osborne (2017) highlighted that automation significantly threatens low-wage and low-education workers. In contrast, this study finds that AI's perceived displacement risk extends to highly educated professionals, aligning with Guliyev's (2023) findings that AI impacts skilled professions differently based on industry context.

Importantly, this study reveals that AI may have a complementary rather than a substitutive relationship with workers' roles. Starkly contrasts the prevailing view in mainstream media and society that AI will replace human jobs. It reveals a more complex and nuanced range of possibilities, suggesting that workers are beginning to recognize the value-added role of AI in their work rather than seeing it solely as a threat. This result aligns with Autor's (2015) historical analysis, which argued that automation typically leads to job transformation rather than elimination.

This study indicates that future policy should consider variables such as gender, age, education, work nature, industry, and occupation to delve more deeply into the intricacies of AI's impact and to develop more targeted countermeasures to confront it effectively. Such an approach will provide a more comprehensive understanding of AI's societal role and inform more effective decision-making.

Statements and declarations

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The sample data used in this manuscript is from the Government of Taiwan, and there is no falsification, forgery, or inappropriate data manipulation in the process of analysis. The sample was made fully public by the Government of Taiwan and did not involve information relating to the informed consent of human participants or animal welfare issues.

CRediT authorship contribution statement

Kuang-Hsien Wang: Writing – original draft, Visualization, Software, Resources, Project administration, Methodology, Investigation, Funding acquisition, Formal analysis, Data curation, Conceptualization. **Wen-Cheng Lu:** Writing – review & editing, Visualization, Validation, Supervision.

Conflict of Interest

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